are present in a fluid matrix called **plasma**. The plasma contains inorganic salts and organic substances. It is a main circulating fluid that helps in the transport of nutrient substances.

Red blood corpuscles (Erythrocytes): The red blood corpuscles are circular, biconcave disc-like cells and **lack nucleus** when mature (mammalian RBC). They contain a respiratory pigment called **haemoglobin** which is involved in the transport of oxygen to tissues.

White blood corpuscles (Leucocytes): They are larger in size, contain **distinct nucleus** and are **colourless**. They are capable of amoeboid movement and play an important role in body's defense mechanism. They engulf or destroy foreign bodies. WBC's are of two types: Granulocytes and Agranulocytes. **Granulocytes** have irregular shaped nuclei and cytoplasmic granules. They include the **neutrophils**, **basophils** and **eosinophils**. **Agranulocytes** lack cytoplasmic granules and include the **lymphocytes** and **monocytes.**

Blood platelets: They are minute, anucleated, fragile fragments of giant bone marrow called **mega karyocytes**. They play an important role in **blood clotting** mechanism.

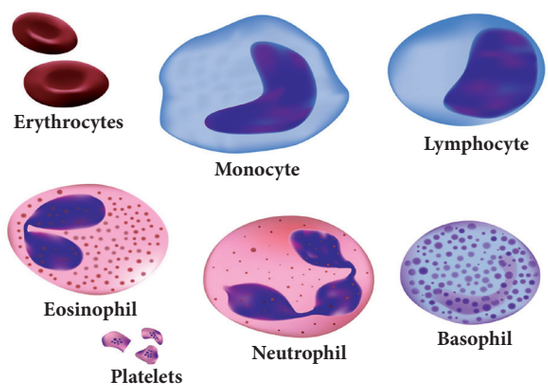


Figure 18.19 Blood cells

b. Lymph

Lymph is a colourless fluid filtered out of the blood capillaries. It consists of plasma and white blood cells. It mainly helps in the exchange of materials between blood and tissue fluids.

18.2.3 Muscular Tissues

Muscular tissues are made of muscle cells and form the major part of contractile tissue. They are composed of numerous **myofibrils**. Each muscle is made up of many long cylindrical fibres arranged parallel to one another.

According to their structure, location and functions there are three main types of muscles: Skeletal muscle (or) striated muscle, Smooth muscle (or) non-striated muscle and Cardiac muscle.

Skeletal muscle: These muscles are attached to the bones and are responsible for the body movements and are called skeletal muscles. They work under our control and are also known as **voluntary muscles**. The muscle fibres are elongated, cylindrical, unbranched with alternating dark and light bands, giving them the **striped** or **striated** appearance. They possess many nuclei (**multinucleate**). For example they occur in the biceps and triceps of arms and undergo rapid contraction.

Smooth muscle: These muscles are spindle shaped with broad middle part and tapering ends. There is a single centrally located nucleus (**uninucleate**). These fibrils do not bear any stripes or striations and hence are called **non-striated**. They are not under the control of our will and so are called **involuntary muscles**. The walls of the internal organs such as the blood vessels, gastric glands, intestinal villi and urinary bladder contain this type of smooth muscle.

Cardiac muscle: It is a special contractile tissue present in the heart. The muscle fibres are **cylindrical**, **branched** and **uninucleate**. The branches join to form a network called as **intercalated disc** which are unique distinguishing features of the cardiac muscles. The contraction of cardiac muscle is involuntary and rhythmic.

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